gn. Remarks

23.09.26

Java Program

Assignment - 01

Q1] What is the extension of a java source file ? What will be the extension of a compiled java file ?

→ The extension of a Java source file is .java

After compilation, java source code is converted into bytecode

The compiled bytecode file has the extension .class

Q2] Given Java source file Hello.java, what command do we use to compile it ?

→ The Java compile javac is used to compile a Java source file

Q3] If the source file hello.java has only one class named Hello, what files will be generated on compiling it ?

→ If the source file contains only one class

```
class Hello
{

}
```

After compilation -

javac hello.java

The compiler generates -

Hello.class

Q4] What is the full name of JDK? What is JRE

→ The Full Name of

JDK - Java Development Kit

JRE - Java Runtime Environment

Q5] How is JDK different from JRE?

JDK	JRE
• Java Development kit	Java Runtime Environment
• Used for developing Java programs	Used for running Java programs
• Contains JRE	Contains JVM & library files
• It contains tools such as javac, javadoc, jdb	It mainly provides runtime environment
• It is required by Java programmers/developer	It is required by users who only run Java applications

Q6] When we install Java 8, do we get a public & a private JRE?

→ In a typical JDK 8 installation, there can be

Public JRE

Private JRE

The public JRE can be used to run Java applications independently

The private JRE is associated with the JDK installation & is used by JDK development tools

07] Where is the private JRE located ?

→ The private JRE is located inside the JDK installation directory

08] When is public JRE used and who uses it ? When is private JRE used ?

→ Public JRE :

It is used to run Java applications

It is useful for users who want to run Java programs but do not need to develop them

It can be installed separately from the JDK

Private JRE :

It is used internally by the JDK development environment

It supports Java development tools & applications.

It is located inside the JDK installation directory

09] Why is Java considered platform independent ? What is Byte code ?

→ Platform independent :

Java is platform independent because Java source code is compiled into bytecode, which can run on any operating system having a suitable JVM

Bytecode :

Bytecode is the intermediate code generated by the Java compiler from a Java source program

It is stored in a .class file & is executed by the JVM.

Q10] Name 5 features of the Java programming language. Explain each in one line.

→ Simple : Java has a simple & easy to understand syntax

Object Oriented : Java programs are based on classes & objects

Platform Independent : Java bytecode can run on different platforms using JVM

Secure : Java provides features such as bytecode verification & controlled memory access

Robust : Java provides exception handling, automatic garbage collection & strong type checking

Q11] Why do we set the PATH of the Java installation folder ?

→ The PATH environment variable is set so that Java commands such as : java, javac, javadoc can be executed from any directory in the command prompt

Q12] If Java is installed in C:\Program Files\Java, what will be the value of JAVA_HOME and what values is added to PATH

→ JAVA_HOME should point to the JDK installation directory, not merely the parent Java folder.

JAVA_HOME -

C:\Program Files\Java\jdk1.8.0_301

PATH :

%JAVA_HOME%\bin

Q13] What are the four pillars of OOPS ? Define each one in one line

→ The four pillars of Object-Oriented Programming are :

① Encapsulation - Wrapping data & methods together into a single unit such as a class

② Abstraction - Showing only the essential features & hiding unnecessary implementation details

③ Inheritance - Acquiring the properties & methods of an existing class into a new class

④ Polymorphism - Allowing the same method or interface to have different forms or behaviors

Q14] Give any non technical ex/ explanation of what each OOPS pillar does

① Encapsulation - ATM : Keeps your account details protected & allows access through operations such as withdrawal & deposit

② Abstraction - Car : A driver uses the steering wheel, brake & accelerator without knowing the internal working of the engine

③ Inheritance - Family : A child can inherit certain characteristics or properties from parents

④ Polymorphism - Person : The same person can perform different roles, such as being a student at college & a customer at a shop

15] What are tokens in the Java programming language?

→ A token is the smallest individual unit of a Java program that has a meaningful purpose to the compiler.

The main types of Java tokens are:

1. Keywords
2. Identifiers
3. Literals
4. Operators
5. Separators

16] What are keywords? How many keywords are there in JAVA?

→ Keywords are reserved words in Java that have a predefined meaning to the compiler. They cannot normally be used as identifiers such as class name or variable names

→ class, public, static, void, int, if, else, for, while.

17] How many keywords were there before Java 5? Which keyword was added in Java 5?

→ Before Java 5, Java had 49 keywords
Java 5 introduced the keywords

→ enum

therefore;

Before Java 5 → 49 Keywords
Java 5 → enum added.
After Java 5 → 50 keywords.

18] Name 5 keywords that you know in Java

→ The 5 keywords are :

1. class
2. Public
3. static
4. void
5. int

19] Write a Java program to print the mesage "Hello Programmer ! , you are welcome ". Also mention commands to compile & run the program with output

→
```
class HelloProgrammer
{
    public static void
    main (String args [])
    {
    System.out.println ("Hello Programmer ! , you are
        welcome ");
    }
}
```

1 → HelloProgrammer.java

2 → Javac HelloProgrammer.java

3 → Java HelloProgrammer

O/p : Hello Programmer ! , you are welcome

20] How many types of whitespaces do we have in the Java programming language?

→ Common whitespace characters used in Java are :-

1. Space
2. Tab (\t)
3. New line / Line feed (\n)
4. Carriage return (\r)
5. Form Feed (\f)

Whitespace is generally used to separate tokens & improve the readability of a Java program

ex : int a = 10 ;

2] Mention two separators that you observed in the code written so far.

→ Two common Java separators are :

① Semicolon - It indicates the end of a statement

ex : int a = 10.

② Braces { } - they indicate the beginning & end of a blocks of code

```
ex : class Hello
     {
       // class body
     }
```